# The Riemann Hypothesis: Spectral Routes and Dead Ends

## Part III of a Five-Part Series

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**Abstract**

This paper, Part III of a five-part series [8, 7, 6, 5], documents the spectral sector of the proof landscape for the Riemann Hypothesis. Two independent conditional reductions are presented:

Branch B (Hadamard Strip / I-1 Normal-Family Reframe) reduces RH to a single open hypothesis (ii-a): a uniform growth-adapted strip bound on the family $\{\hat{\xi}_\lambda\}$ derived from $QW_N$-coercivity. The multiplicative canonical form $A_\lambda(\delta) = A_\Xi(\delta)\big(1 + \kappa(\lambda)\delta^2\big)$ with saturating $\kappa_\infty \in [1.6, 1.8] \times 10^{-3}$ is established empirically, yielding Theorem B-1 (conditional): $A_\lambda(\delta) \leq 1.0063$ under uniform log-curvature control.

Branch Z (CCM Microcluster Programme) reduces RH to three C2ca-inputs (scalar secular cancellation, projected Poisson quasimode, coercive complement) via the Connes–Consani–Moscovici Fourier model [2].

Both branches share Wall (ii) as the common analytical obstacle. Fifteen excluded spectral and CCM paths are catalogued, establishing the productive routes as the only viable approaches in the current landscape.

**Series status (v3.0, 2026-05-26).** The series is restructured from a trilogy to a five-part series; the conditional reduction status from v2.2 is unchanged. New in this part: the multiplicative canonical form (Section 2.4), Theorem B-1 (conditional uniform strip bound with $A_\lambda \leq 1.0063$, Theorem 2.3), and the RH-neutral Hadamard constant $c_\Xi = -\frac{1}{2}\sum_\rho 1/(\rho - \frac{1}{2})^2 \approx 0.0231$ (Section 4).

**MSC 2020:** 11M26, 11M36, 47A75, 30D15

**Keywords:** Riemann Hypothesis, spectral routes, normal family, Hadamard strip bound, CCM microcluster, conditional reduction, excluded paths

# Contents

# 1 Introduction: The Spectral Sector

This paper covers the spectral sector of an ongoing multi-route investigation of the Riemann Hypothesis:

**Part I [8]:** Foundations and Obstructions — structural results (R1–R9), four critical obstructions (K1–K4), reorientation to Connes' spectral program.

**Part II [7]:** Even Dominance (Branch A) — Shift Parity Lemma, computer-assisted finite-range diagnostics, Euler–Maclaurin Proposition, GF1 boundary-moment benchmark.

**Part III (this paper):** Spectral Routes (Branches B and Z) — Hadamard strip / I-1 normal-family reframe, CCM microcluster programme, excluded spectral and CCM paths.

**Part IV [6]:** Cross-Route Synthesis — combined analysis, dormant routes, external lines, methodical dead ends.

**Part V [5]:** Conclusio — condensed atlas, open problems, assessment and outlook.

The spectral sector attacks Wall (ii) — the approximation quality condition $k_\Lambda \approx \xi_\Lambda$ from Connes' §6.6(b) [1] — through two structurally distinct mechanisms. Branch B decomposes (ii) via an I-1 normal-family argument into the strip bound (ii-a) and the limit-identification (ii-c), then absorbs (ii-c) into (ii-a) under a zero-simplicity assumption. Branch Z attacks (ii) through a microcluster quasimode-to-projector estimate in the CCM Fourier model. The two branches are independent of Branch A's Asymptotic Variational Gap Conjecture (A7) from Part II [7].

Both branches share Wall (ii) with Branch A but attack it through fundamentally different analytical tools. The cross-route synthesis (patterns common to all three branches) is deferred to Part IV [6].

# 2 Branch B: Hadamard Strip / I-1 Normal-Family

Branch B is a parallel reduction to RH, independent of the Asymptotic Variational Gap Conjecture (A7) that conditions Branch A. Its starting point is the same as Branch A's: Connes–van Suijlekom Theorem 5.6 [3] provides the finite-$N$ error bound $\mathrm{err}_\lambda \leq Ce^{-c\lambda}$ for each individual Riemann zero, verified numerically at $\lesssim 10^{-25\lambda}$ for $\geq 15$ zeros. Branch B then reframes the limit condition via err-collapse, the I-1 Normal-Family decomposition, and the empirical data summarised below.

## 2.1 The Reduction Chain (B1–B11)

| Step | Statement | Status | Source |
|---|---|---|---|
| B1 | Connes' Theorem 6.1 / even dominance ⇒ zeros of $\hat{\eta}$ real (= A1) | proven | [3, 1] |
| B2 | Condition (i): real zeros of $\hat{\xi}_\lambda$ at finite $N$ (CvS Thm 5.6 / 1-dim even kernel) | proven at every finite $N$ | companion proof notes |
| B3 | err-collapse equivalence: err $\to 0 \iff$ (i) + (ii) | rigorous reduction | companion proof notes §1 |
| B4 | err-collapse empirics: err $\sim 10^{-25\lambda}$ for $\geq$ 15 zeros, 4 $\lambda$-points | numerical evidence | companion proof notes §9 |
| B5 | I-1 Normal-Family Reframe: (ii) = (ii-a) $\wedge$ (ii-c) via Montel–Vitali compactness | rigorous decomposition | §E.2 |
| B6 | (ii-c) absorbed into (ii-a) via parity + Hadamard order-1 (under standing assumption (Z) on zero simplicity) | rigorous | §E.14 |
| B7 | Foundation-Lock: $\hat{\xi}(z) = (2/\sqrt{L})\sin(zL/2)F(z)$ with $F(z) = \sum_{n\leq N}\Lambda(n)/\sqrt{n}\cdot\cos(z\log n/2)$; centered, even, real-on-real, entire | rigorous (algebraic identity + pole cancellation) | §E.11, §E.15 |
| B8 | Strip amplification flat at converged precision: $A_\lambda(0.4) \approx 1.004$ across $\lambda \in \{3, 5, 7, 9, 11, 13, 15\}$ | numerical evidence, dps $\approx$ 30–50 $\cdot\,\lambda$ | §E.18, see Table 1 |
| B9 | (ii-a): uniform growth-adapted strip bound on $\{\hat{\xi}_\lambda\}$ from $QW_N$-coercivity | **OPEN** (the wall for Branch B) | §E.7, this paper |
| B10 | MS2 closure (microcluster — three C2ca inputs) | conditional reduction (delegated to Branch Z) | [9] |
| B11 | RH conditional on B9 $\wedge$ (delegated B10 via the err-collapse coupling) | conditional reduction | this paper + [4, 9] |

## 2.2 The I-1 Normal-Family Reframe and the Foundation-Lock

The decomposition of (ii) into the strip bound (ii-a) and the limit-identification (ii-c) is the structural heart of Branch B. Montel–Vitali compactness on the normal family $\{\hat{\xi}_\lambda\}$ reduces local uniform convergence to two sub-conditions; the parity of $\mathrm{QW}_N$ (real-symmetric, even-parity projection ⇒ $\hat{\xi}_\lambda$ is even and real-on-real) together with a Hadamard order-one argument forces the Bohr–Carathéodory phase factor to be identically 1 under standing assumption (Z) (zeros simple). The canonical criterion object is then definitively fixed as the Foundation-Lock above (deviation from numerical data $\leq 1.9\times10^{-52}$ at $\lambda = 5, 7, \mathrm{dps} = 50$).

Branch B therefore reduces *entirely to the single open condition (ii-a)*: a rigorous uniform growth-adapted strip bound derived from $\mathrm{QW}_N$-coercivity. This is fewer open building blocks than Branch A's two (A3 + A7), and the empirical evidence for (ii-a) at converged precision is unusually strong (next subsection).

## 2.3 Empirical Convergence (Mac Studio, converged dps)

Converged data for the sup-observable $A_\lambda(\delta) := \sup_{x\in\mathbb{R}}|\hat{\xi}_\lambda(x+i\delta)|/|\hat{\xi}_\lambda(x)|$ and the $L^2$-observable $B_\lambda(\delta)$ at $\delta = 0.4$, obtained on a dedicated compute node (Mac Studio, $\mathrm{dps}_{\min} \approx 30\lambda$–$50\lambda$ to control cluster near-degeneracy; two distinct dps values bit-identical at $\lambda = 11, 13, 15$):

The near-flatness of $A_\lambda(0.4)$ shows that the family $\{\hat{\xi}_\lambda\}$ is strongly sub-generic, making (ii-a) empirically plausible. The $L^2$-observable grows as $B_\lambda(0.4) \approx 0.68\lambda^{0.4}$ on the plateau, consistent with polynomial sub-generic behaviour. Earlier runs at dps = 30

| $\lambda$ | $\sup\lvert\hat{\xi}_\lambda\rvert_{\mathbb{R}}$ | $A_\lambda(0.4)$ | $B_\lambda(0.4)$ | $B_\lambda(0.4)/\lambda^{0.4}$ |
|---|---|---|---|---|
| 3 | 0.858 | 1.0033 | 1.151 | 0.74 |
| 5 | 0.879 | 1.0037 | 1.340 | 0.70 |
| 7 | 0.886 | 1.0038 | 1.505 | 0.69 |
| 9 | 0.890 | 1.0039 | 1.649 | 0.68 |
| 11 | 0.890 | 1.0039 | 1.778 | 0.68 |
| 13 | 0.891 | 1.0039 | 1.895 | 0.68 |
| 15 | 0.892 | 1.00394 | 2.003 | 0.68 |

Table 1: Converged empirics for Branch B observables at $\delta = 0.4$ (companion proof notes, §E.18). The sup-observable $A_\lambda(0.4)$ is essentially flat at $\approx 1.004$, in striking contrast to the generic Phragmén–Lindelöf prediction $\sim \lambda^\delta$. The $B_\lambda$-ratio decreases from 0.74 at $\lambda = 3$ to a stable plateau at 0.68 for $\lambda \geq 9$.

(§E.16–E.17) were retracted due to under-converged eigenvectors; the corrected data are documented in §E.18 of the companion notes [4].

## 2.4 Multiplicative Canonical Form

The strip-amplification data (Table 1) admit a factored representation in terms of the completed Riemann $\xi$-function. Define the *Hadamard baseline*

$$A_\Xi(\delta) := \frac{\sup_{x\in\mathbb{R}} |\Xi(x + i\delta)|}{\sup_{x\in\mathbb{R}} |\Xi(x)|}, \tag{1}$$

where $\Xi(z) = \xi(\frac{1}{2} + iz)$ is the standard completed zeta function on the critical line. By the Hadamard product, $\log A_\Xi(\delta) \sim c_\Xi\,\delta^2$ for small $\delta$, where the Hadamard curvature constant is (see Section 4)

$$c_\Xi = -\frac{1}{2}\sum_\rho \frac{1}{(\rho - \frac{1}{2})^2} \approx 0.0231, \tag{2}$$

summing over all non-trivial zeros $\rho$ of $\zeta(s)$. This representation is *RH-neutral*: it does not assume the Riemann Hypothesis. Under RH, the sum specialises to $\sum_{k\geq 1} 1/\gamma_k^2$ with $\gamma_k = \Im(\rho_k)$.

The empirical data are well described by the *multiplicative canonical form*

$$A_\lambda(\delta) = A_\Xi(\delta)\,\bigl(1 + \kappa(\lambda)\,\delta^2\bigr), \tag{3}$$

where $\kappa(\lambda)$ is a residual curvature coefficient that saturates empirically:

$$\kappa_\infty := \lim_{\lambda\to\infty} \kappa(\lambda) \in [1.6, 1.8] \times 10^{-3} \tag{4}$$

(six $\lambda$-points including $\lambda = 20$, with $\kappa(20) = 1.527 \times 10^{-3}$ and stable deviation from the factored form). The saturation is consistent with the Foundation-Lock structure: $\hat{\xi}_\lambda$ converges to $\Xi$ in a growth-controlled sense, and $\kappa(\lambda)$ measures the residual finite-$\lambda$ strip-curvature excess.

*Remark* 2.1 (Pointwise identity, not uniform bound). Equation (3) is a pointwise algebraic decomposition at each $\delta$, not a uniform strip bound. The sup over $x$ is taken separately on each side. Converting this to a *uniform* $A_\lambda(\delta) \leq C$ for all $\lambda$ and $\delta \in [0, \frac{1}{2})$ requires the separate log-curvature control of Theorem B-1 below.

## 2.5 Theorem B-1: Conditional Uniform Strip Bound

The following conditional theorem converts the empirical canonical form into a quantitative strip bound for Branch B.

**Definition 2.2.** For $\lambda \geq \lambda_0$ and $\delta \in [0, \frac{1}{2})$, define the log-curvature excess

$$H_\lambda(\delta) := \log \frac{A_\lambda(\delta)}{A_\Xi(\delta)}.$$

**Theorem 2.3** (Conditional uniform strip bound)**.** *Suppose there exist constants $\lambda_0$ and $K \leq 1.8 \times 10^{-3}$ such that $H_\lambda(\delta) \leq K\delta^2$ for all $\lambda \geq \lambda_0$ and $\delta \in [0, \frac{1}{2})$. Then*

$$A_\lambda(\delta) \leq B_\Xi \cdot \exp\bigl(K/4\bigr) \leq 1.0063, \tag{5}$$

*where $B_\Xi := \xi(0)/\xi(\frac{1}{2}) = 1.005791795350375\ldots$ is the analytical upper bound for $A_\Xi$ on $[0, \frac{1}{2})$.*

*Proof sketch.* By the hypothesis, $A_\lambda(\delta) \leq A_\Xi(\delta)\, e^{K\delta^2}$. The Hadamard product for $\Xi$ gives $A_\Xi(\delta) \leq B_\Xi$ for $\delta \in [0, \frac{1}{2})$, where $B_\Xi = \Xi(0)/\Xi(i/2) = \xi(0)/\xi(\frac{1}{2})$ is attained at $\delta = \frac{1}{2}$ (limit value). The exponential factor is maximised at $\delta = \frac{1}{2}$: $e^{K/4} \leq e^{1.8\times10^{-3}/4} \leq 1.00045$. Hence $A_\lambda(\delta) \leq 1.005792 \times 1.00045 \leq 1.0063$. □

*Remark* 2.4 (Open proof slot)*.* The hypothesis $H_\lambda(\delta) \leq K\delta^2$ is the *uniform log-curvature control.* Deriving it rigorously from the Foundation-Lock structure

$$\hat{\xi}(z) = \frac{2}{\sqrt{L}} \sin\bigl(\tfrac{zL}{2}\bigr)\, F(z)$$

is an isolated open proof slot for Branch B. The empirical data (Table 1) satisfy this condition with $K$ in the 1.6–1.8 $\times 10^{-3}$ saturation range, matching the conservative theorem bound.

# 3 Branch Z: CCM Microcluster Programme

Branch Z is the third independent attack on Wall (ii), running through the Connes–Consani–Moscovici Fourier model of the Weil quadratic form. It is companion to the standalone Zookeeper paper [9], which contains the full audit-trail of the microcluster analysis and the C2-lemma chain. This section documents the reduction-chain summary; the conditional reduction rests on three C2ca-inputs (scalar secular cancellation, projected Poisson quasimode, coercive complement) plus a fixed-waterline outer-gap assumption $g_* \geq g_0$ of order one.

## 3.1 The Reduction Chain (B1–B11, CCM-Branch)

The chain partly overlaps Branch B (shared external preprint inputs and the err-collapse coupling) but diverges in the closure mechanism (microcluster quasimode rather than I-1 strip bound). Labels B1–B11 are reused with the CCM-Branch reading.

| Step | Statement (CCM-Branch reading) | Status | Source |
|---|---|---|---|
| B1 | $\mathrm{W02}_{\mathrm{even}}$ is rank-one (pole at $s = 1$): $\tilde{C} = +4.864$, nullspace dim 30 at $(\lambda, N) = (3, 30)$ | proven | `cluster_mechanism_analysis.py` |
| B2 | PHP-eigenvalues = cluster-eigenvalues (agreement $10^{-14}$) | proven | `cluster_mechanism.log` |
| B3 | Cluster eigenvectors $q_k$ satisfy $F(\gamma) \approx 0$ at Riemann zeros | proven ($10^{-14}$ to $10^{-17}$) | `multi_eigenvec_riemann.log` |
| B4 | 15-digit cancellation: pole-term + null-term $\approx 0$ | proven | `weil_positivity_analysis.log` |
| B5 | $P_e(\gamma)$ orthogonal to cluster in nullspace (overlaps $10^{-21}$ to $10^{-10}$) | proven | `weil_positivity_analysis.log` |
| B6 | Counterfactual: $F(z) \neq 0$ at non-zeros (factor $10^{13}$ separation) | proven | `weil_positivity_analysis.log` |
| B7 | Relative positivity $w_{\min}/\tilde{C} \to 0$ for $\lambda \to \infty$ | proven (numerical) | `lambda_positivity_convergence.log` |
| B8 | $w_{\min}$ converges at fixed $N$; bounded asymptote $\approx -2.36$ | proven | `lambda_positivity_convergence.log` |
| B9 | N-scaled convergence: $w_{\min}$ independent of $N$, $w_{\min}/\tilde{C} \to 0$ | proven (partial), server run continues | `n_scaled_convergence.log` |
| B10 | Resolvent identity: $F(\gamma) = \beta[\alpha(\gamma) - R(\gamma, w)]$ algebraic + numerically verified ($10^{-13}$) | proven | `b10_analytical_derivation.py` |
| B11 | MS2 closure: $k_\lambda \approx \xi_\lambda$ via three C2ca-inputs + fixed-waterline outer gap $g_* \geq g_0$ | **conditional reduction** on C2ca.1 + C2ca.2 + C2ca.5 + $g_*$-assumption | [9], C2ca chain |

## 3.2 The Three C2ca-Inputs

The Zookeeper paper [9] reduces MS2 (Connes' §6.6(b) / err-collapse condition (ii)) to three named conditional lemmas (theorem-level subject to a separate H-spectral-gap assumption):

- **C2ca.1 Scalar Secular Cancellation:** algebraic secular identity proven; uniform decay $s_\lambda \to 0$ proven only under a separate Galerkin-truncation control. Numerical evidence: $|\mu/\alpha| \approx 4$–$5 \times 10^{-7}$ stable across $\lambda = 3, 5, 7$.
- **C2ca.2 Projected Poisson Quasimode:** transfer formula $h = (1/n)P_0\Pi(E + B)$ proven; uniform analytical control of $P_0$-leakage and boundary decay open. Numerical evidence: $p_\lambda \approx 2$–$3 \times 10^{-6}$.
- **C2ca.5 Coercive Complement:** finite Min–Max identity for a prescribed spectral window proven; the uniform order-one gap $g_*$ is not derived from Cauchy-interlacing alone — it requires a separate H-spectral gap or fixed-waterline / slush-collar control (replaced by C2cf.1 *Fixed-Waterline Quasimode Projection*: in the finite self-adjoint Galerkin model, for any pre-chosen $g_0 > 0$, $\|(I - P_{\lambda,g_0})k\| \leq R_0/g_0$).

These three inputs plus the fixed-waterline assumption replace the earlier mass-adaptive gap rhetoric (which was diagnosed as circular against MS2, see Section 7).

## 3.3 Empirical Convergence and the Effective Microcluster

The mass-based effective cluster gap $g_{\mathrm{eff}}(\varepsilon) := u_{J+1} - u_0$ with $J_\varepsilon := \min\{J : M(J) \geq 1-\varepsilon\}$ stabilises at $g_{\mathrm{eff}}(10^{-12}) \approx 5$–$6$ (order one, $\lambda$-independent), giving $R_0/g_{\mathrm{eff}} \approx 5 \times 10^{-7}$. The earlier fixed-tolerance gap was unstable (cut inside the genuine cluster); the mass-based replacement is stable and macroscopic. See Zookeeper C2by/C2bz for the full analysis [9].

# 4 Spectral Convergence Constants

Both spectral branches produce characteristic constants whose identification and saturation provide evidence for the conditional reductions. We collect them here for comparison; the cross-route synthesis with Branch A's constant $C^* = 64\pi^2/3$ (Part II [7], § GF1) is deferred to Part IV [6].

## 4.1 Hadamard Curvature Constant $c_\Xi$ (RH-neutral)

The Hadamard product for the completed Riemann $\xi$-function gives

$$\log \frac{|\Xi(it)|}{|\Xi(0)|} = \sum_\rho \log\left|1 - \frac{-t^2}{(\rho-\frac{1}{2})^2}\right| \sim -\frac{t^2}{2}\sum_\rho \frac{1}{(\rho-\frac{1}{2})^2} + O(t^4),$$

whence

$$c_\Xi = -\frac{1}{2}\sum_\rho \frac{1}{(\rho-\frac{1}{2})^2} \approx 0.0231. \tag{6}$$

This is the leading quadratic curvature of $\log A_\Xi(\delta)$ near $\delta = 0$. The representation (6) sums over *all* non-trivial zeros and is *RH-neutral*. Under RH, $\rho = \frac{1}{2} + i\gamma_k$ and the sum specialises to $\sum_{k\geq 1} 1/\gamma_k^2$.

## 4.2 Residual Curvature $\kappa_\infty$ (Branch B)

The multiplicative canonical form (3) decomposes $A_\lambda$ into the Hadamard baseline $A_\Xi$ and a residual factor $1 + \kappa(\lambda)\delta^2$. Empirically, $\kappa(\lambda)$ saturates to $\kappa_\infty \in [1.6, 1.8] \times 10^{-3}$ across $\lambda = 3, 5, 7, 9, 11, 13, 15, 20$, with $\kappa(20) = 1.527 \times 10^{-3}$. This saturation is consistent with the Foundation-Lock convergence: as $\lambda \to \infty$, $\hat{\xi}_\lambda \to \Xi$ in the relevant topology, and the residual curvature measures the finite-$\lambda$ correction.

## 4.3 Hecke-Normalised Gap $g_0(m)$ (Branch Z)

For the $m$-th Riemann zero $\gamma_m$, the classical Montgomery–Odlyzko pair-correlation heuristic [10] suggests a local mean zero spacing of $2\pi/\log\gamma_m$. The effective microcluster gap $g_{\text{eff}}$ of Branch Z (Section 3.3) is consistent with this scaling: $g_0(m) \approx 2\pi/\log\gamma_m$.

| Branch | Constant | Value | Status |
|---|---|---|---|
| B | $c_\Xi$ (Hadamard curvature) | $\approx 0.0231$ | analytical (RH-neutral) |
| B | $\kappa_\infty$ (residual curvature) | $\in [1.6, 1.8] \times 10^{-3}$ | empirical (saturated) |
| B | $B_\Xi = \xi(0)/\xi(1/2)$ | $1.00579\ldots$ | analytical |
| Z | $g_0(m) \approx 2\pi/\log\gamma_m$ | varies | classical (Montgomery–Odlyzko) |

# 5 Common Wall and Spectral-Sector Outlook

The three branches share Wall (ii) as the common analytical obstacle, but attack it through structurally different mechanisms. The following comparison table summarises the conditional reductions.

| Branch | Reduction chain | Open hypothesis (the wall here) | Status |
|---|---|---|---|
| A — Even Dominance | A1 + A2 + A7 ⇒ RH | A7 (Asymp. Variational Gap Conj.) + A3 (Odd-sector lower bound) | conditional reduction |
| B — I-1 Normal-Family | B1 + B5 + B9 ⇒ RH | (ii-a) (uniform strip bound from $QW_N$-coercivity) | conditional reduction |
| Z — CCM Microcluster | B1 + C2ca + fixed-waterline ⇒ RH | C2ca.1 + C2ca.2 + C2ca.5 + $g_* \geq g_0$ | conditional reduction |

**Shared inputs.** A1 = B1 (both rest on Connes–vS Thm 5.6 [3]). The Foundation-Lock $\hat{\xi}(z) = (2/\sqrt{L})\sin(zL/2)F(z)$ from Branch B is the canonical criterion object used in Branch Z as well.
**Distinct mechanisms.** Branch A attacks Wall (ii) via eigenvalue ordering $\lambda_1^+ < \lambda_1^-$ (Part II [7]). Branch B attacks via a strip-uniform Hurwitz-style normal-family argument (this paper). Branch Z attacks via a microcluster quasimode-to-projector estimate (this paper + Zookeeper [9]).
**Closest to conditional closure.** Branch B currently has the smallest open surface (a single named hypothesis (ii-a)) together with strongly sub-generic empirical evidence ($A_\lambda(0.4) \approx 1.004$ flat over seven converged $\lambda$-points, Theorem B-1 giving $A_\lambda \leq 1.0063$ under the log-curvature hypothesis). Branches A and Z remain conditional reductions in the multi-gap sense (two or three open inputs respectively). The title of the present series follows the conservative convention: "Atlas" applies as long as no single branch has reached conditional-proof level under the audit protocol of the project `DECISION.md`.
**Promotion protocol and future scenarios** are discussed in Part V (Section 9 gives a brief forward reference).

# 6 Excluded Spectral Paths

The post-v2.0 spectral programme (Connes–van Suijlekom reformulation, I-1 normal-family reframe, err-collapse diagnostic) closed several routes. These are documented here for the audit trail; they do not affect the main conditional reduction.

- **Eigenvalue-ordering via a uniform spectral gap.** The plan to lift $\lambda_{\min}(W^+ - W^-) < 0$ to the eigenvalue inequality $\lambda_1^+ < \lambda_1^-$ via Kato–Temple, Davis–Kahan, or Lehmann gap estimates fails because the relevant minimum eigenvector of $QW_N$ sits in a near-degenerate cluster (even/odd split $\sim 10^{-65}$ at $\lambda = 5$, $\sim 10^{-84}$ at $\lambda = 7$, super-exponentially collapsing with $\lambda$). No uniform spectral gap is available.

- **Localization hypothesis for the strip bound (ii-a).** A natural plan was to bound $|\hat{\xi}_\lambda(x + i\delta)|$ via localization assuming $\xi_\lambda$ is concentrated near $t = 0$. This is falsified: $\xi_\lambda$ is genuinely edge-concentrated ($r_{99} \approx L/2$, $|\xi_\lambda(0)| \approx 0$). A growth-adapted normalisation $\Phi$ is required.

- **Edge-profile factorisation for (ii-a).** The exact identity

$$F_\lambda(z) = 2\cos(zL/2)\, C_\lambda(z) + 2\sin(zL/2)\, S_\lambda(z)$$

  isolates the explicit $\lambda^\delta$-growth modulation. The edge route yields no type gain because the eigenfunction mass is spread ($r_{50} \approx 1$ uniformly, but $r_{99} \approx L/2$).

- **Literal MS1 (cluster ladder).** The conservative form of MS1 — uniformly simple splitting of the minimum eigenvalue of $QW_N$ at every $(\lambda, N)$ — fails: the splitting collapses super-exponentially with $\lambda$. Only a cluster/quotient reformulation has structural traction.

- **Strip-amplification at** dps $= 30$ **(retracted).** Early measurements at dps $= 30$ suggested sub-generic growth and non-monotone instability. At converged precision (dps $\approx 30$–$50 \cdot \lambda$), the picture is qualitatively different: $A_\lambda(0.4) \approx 1.004$ is flat. The methodological lesson (dps requirement scales linearly with $\lambda$ for cluster-near-degenerate eigenvectors) is itself substantive.

- **Raw-$L^2$ Hurwitz passage (falsified 2026-05-15).** The plan to derive the Hurwitz strip-uniform convergence directly from a raw-$L^2$ tail estimate fails on two fronts: (a) the archimedean tail has norm $O(\lambda^{-1/2})$ — exactly boundary exponent $\alpha = 0$, no margin; (b) the prime-Dirac tail diverges in the $m = 1$ branch. Negative-Sobolev does not rescue this. The diagnostic identifies operator-norm topology (i.e. Branch B) as the only viable replacement. The Connes–Hurwitz-Passage project rescoped on 2026-05-17 to a purely expositorial role (`Geiger2026hurwitz`, in preparation).

*Remark* 6.1 (Common pattern). These six exclusions share a structural pattern: the minimum-eigenvector cluster of $QW_N$ is the source of the technical difficulty. Methods that assume or require a uniform gap, a thin-localised eigenfunction, or a generic function-class behaviour all fail. Productive routes work *with* the cluster (waterline / cluster-quotient reformulations, growth-adapted normalisations) rather than against it.

# 7 Excluded CCM Paths (Branch Z Heritage)

The CCM microcluster programme accumulated its own catalogue of dead routes. They do not affect the Branch-Z conditional reduction.

- **Mass-based microcluster definition.** Defining the microcluster via a mass criterion is circular against MS2. Replaced in Zookeeper v1.2+ by a *spectral* microcluster (low-energy eigenspace bounded by an outer gap $g_*$ of order one).

- **Naive Hurwitz passage.** A direct Hurwitz application without approximation-quality control yields statements too weak for the err-collapse closure. Replaced by a Slepian–Pollak-/PSWF-based passage.

- **Anchor-perturbation / Banach contraction (C2br′, falsified).** A Banach-norm contraction $C_{\max} < M$ fails by factor 55–7000 on all twelve tested configurations.

- **Projected frontier cancellation $S_{bulk} \approx -S_{bd}$ (C2bs, falsified).** Both contributions have the same sign and are both of size $\sim 10^{-7}$. The true cancellation mechanism sits in the eigenvalue equation $(A - w_0)k \approx 0$.

- **Uniform mass criterion C2ak.** The total off-cluster mass bound diverges with $\lambda$ ($6.68 \to 15.3 \to 1865$ across $\lambda = 3, 5, 7$) because it discards internal cancellation.

- **Canonical local geometry C2al (directional defect).** The geometric inequality has the wrong direction — the correct form is $g_R/d_m \leq T_m^\beta/\beta_0$ (lower bound on $d/g$, not upper).

- **Cluster selection via $\Delta$-projection.** Simple $\delta_N$-projection is not a robust cluster selector: at $\lambda = 5, N = 55$ `best_zero=k=16` while the other two select $k = 17$.

- **Log-concavity / detailed balance (C2cc H1/H2/H4).** The hypotheses that $\beta_n$ or $|b_n|$ are log-concave (H1), that the spectral residue chain has a detailed-balance structure (H2), or that a Mertens-anchor applies (H4) were tested at $\lambda = 3, N = 30, \text{dps} = 50$ and falsified. Structural cause: arithmetic (prime-driven) oscillation.

- **Direct Frontier-Dominance v2.1 (Killer 0, 2026-05-14).** The v2.1 argument used a common Rayleigh test vector to prove $\lambda_{\min}(D_{tot}) < 0$ and then identified $\Delta_{low} = \lambda_{\min}(D_{tot})$ with $\Delta = \lambda_1^+ - \lambda_1^-$. The reviewer counterexample $A = \text{diag}(0, 10), B = \text{diag}(5, -5)$ shows this does *not* follow. Even dominance for asymptotic $\lambda$ remains conditional on a separate odd-sector lower bound. (This item is Even-sector-specific and documented in detail in Part II [7]; it is listed here because the falsification also restructured the comparative analysis of Branches B and Z.)

*Remark* 7.1 (CCM-pattern: cancellation, not magnitude). The CCM-side dead routes share a different pattern from the spectral side: each tries to bound a quantity by its individual-term magnitude, and each is defeated by large internal cancellation. The living routes explicitly factor the cancellation through projector identities or scalar secular equations, never through simple norm domination.

# 8 Methodical Dead Ends

A small number of methodical routes were tested on the spectral side and ruled out on structural grounds.

- **Dobrushin–Zegarlinski uniqueness → RH-CAP.** Dobrushin–Zegarlinski uniqueness works only on lattices with localised Gibbs measures. RH has neither; the absence of multiplicative independence enforces a non-local prime-coupling structure that breaks the Dobrushin contraction setting.

- **Sturm node-counting (naive).** Classical Sturm–Liouville node-counting for the eigenfunctions of $QW_N$ fails because the operator is finite-dimensional and the even/odd spectral gap is sub-exponentially small. Node-counting provides no information at the relevant precision scale.

- **Naive Abel summation for far-tail.** Applying Abel summation to the far-tail prime sums $\sum_{p>\lambda} (\log p)^k / p^s$ without controlling the arithmetic oscillation of $\theta(x) - x$ leads to divergent error terms. The PNT transfer lemma (Part II) shows how to do this correctly.

- **Naive superdeterminant form for Möbius-$\mathbb{Z}_2$.** The formal identity $\text{sdet}(I - L_s) = \det(I - L_s^+) / \det(I - L_s^-) = 1/\zeta(s)$ is algebraically correct for finite prime sets but does not converge for the full prime set at $\Re(s) = \frac{1}{2}$ (trace-class obstruction K4 from Part I).

- **Friedrichs + Petersson as Hilbert–Pólya lever.** The plan to use Friedrichs extension of the Petersson inner product to construct a self-adjoint realisation whose spectrum encodes the Riemann zeros fails: the relevant bilinear form is not semi-bounded on the domain needed to capture the critical-line zeros.

# 9 Summary

| Contribution | Status | Location |
|---|---|---|
| I-1 Normal-Family Reframe: (ii) $\to$ (ii-a) + (ii-c) | rigorous | Section 2.2 |
| (ii-c) absorbed into (ii-a) under (Z) | rigorous | Section 2.2 |
| Foundation-Lock: canonical criterion object | rigorous | Section 2.2 |
| Converged empirics: $A_\lambda(0.4) \approx 1.004$ (7 points) | numerical evidence | Table 1 |
| Multiplicative canonical form | empirical (0.012% fit) | Section 2.4 |
| Theorem B-1: $A_\lambda \leq 1.0063$ (conditional) | conditional | Theorem 2.3 |
| $c_\Xi$ Hadamard curvature (RH-neutral) | analytical | Section 4 |
| CCM: three C2ca-inputs identified | conditional reduction | Section 3.2 |
| CCM: effective microcluster gap $g_{\text{eff}} \approx 5$–$6$ | numerical evidence | Section 3.3 |
| 6 excluded spectral paths + 9 excluded CCM paths | documented | Sections 6 and 7 |
| 5 methodical dead ends | documented | Section 8 |

The spectral sector contributes two independent conditional reductions of RH, each with a well-characterised open surface. Branch B has the smallest open surface (a single hypothesis (ii-a)) and the strongest empirical support ($A_\lambda \leq 1.0063$ conditional on uniform log-curvature control). The cross-route synthesis across all three branches is developed in Part IV [6]; the condensed atlas and outlook in Part V [5].

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# References

[1] Alain Connes. The Riemann hypothesis: Past, present and a letter through time. *arXiv preprint*, 2026. Survey on the Riemann Hypothesis: past, present, and a letter to Riemann using only pre-1860 mathematics.

[2] Alain Connes, Caterina Consani, and Henri Moscovici. Zeta zeros and prolate wave operators. *Annals of Functional Analysis*, 15(87), 2024. Jacobi matrix representation and prolate–zeta connection.

[3] Alain Connes and Walter D. van Suijlekom. Quadratic forms, real zeros and echoes of the spectral action. *Communications in Mathematical Physics*, 406(12), 2025. Quadratic forms, real zeros, and connections to the spectral action.

[4] Lukas Geiger. A conditional reduction of the Riemann hypothesis to even dominance of the Weil quadratic form. Zenodo, 2026. Standalone extract from the Landscape series; v1.8 record DOI 10.5281/zenodo.20396370, concept DOI 10.5281/zenodo.19764771.

[5] Lukas Geiger. The Riemann hypothesis: Conclusio. Preprint, 2026. Part V of a five-part series (in preparation).

[6] Lukas Geiger. The Riemann hypothesis: Cross-route synthesis. Preprint, 2026. Part IV of a five-part series (in preparation).

[7] Lukas Geiger. The Riemann hypothesis: Even dominance of the Weil quadratic form. Preprint, 2026. Part II of a five-part series.

[8] Lukas Geiger. The Riemann hypothesis: Foundations, obstructions, and reorientation. Preprint, 2026. Part I of a five-part series.

[9] Lukas Geiger. The spectral zookeeper: A conditional reduction toward the Riemann hypothesis via spectral microcluster closure in the Connes–Consani–Moscovici framework. Zenodo, 2026. Companion CCM-route paper; v1.4 record DOI 10.5281/zenodo.20396399, concept DOI 10.5281/zenodo.19673126.

[10] Hugh L. Montgomery. The pair correlation of zeros of the zeta function. *Proceedings of Symposia in Pure Mathematics*, 24:181–193, 1973.